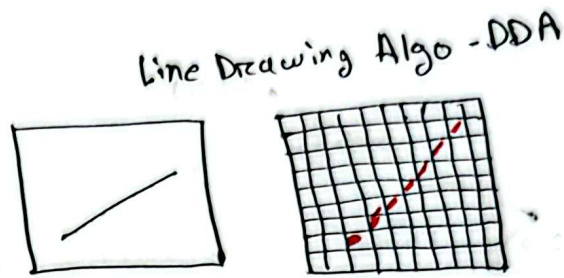
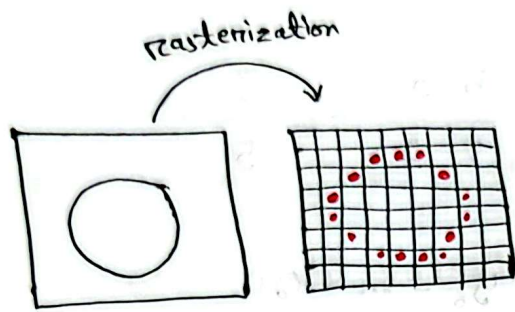


Computer Graphics

Scan Conversion Algorithms



Considerations:

(i) The line has to look good
(Avoid jaggies)

* (ii) It has to be lightening fast!

Naive
DDA
MPL (best)

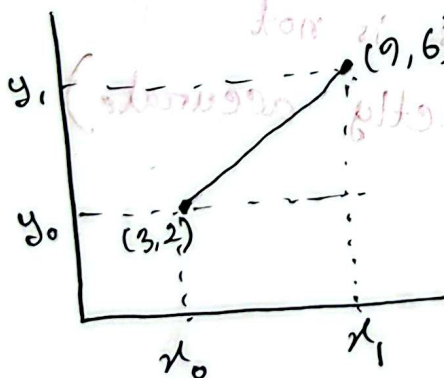
faster

line equations:

$$y = mx + b$$

$$m = \frac{y_1 - y_0}{x_1 - x_0} = \frac{\Delta y}{\Delta x}$$

$$b = y_0 - mx_0$$

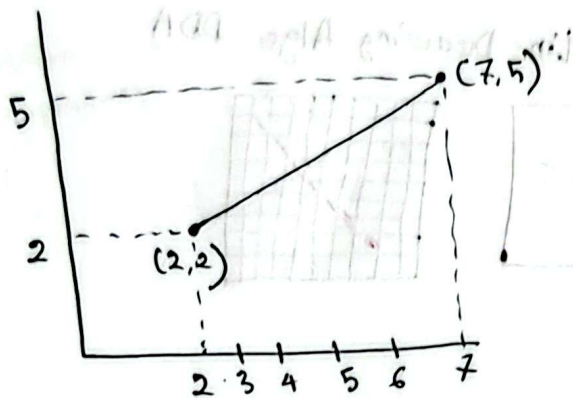


$$m = \frac{6 - 2}{9 - 3} = \frac{4}{6} = \frac{2}{3}$$

$$b = 2 - \frac{2}{3} \times 3 = 2 - 2 = 0$$

A very simple solution:

Naive solution



$$m = \frac{5-2}{7-2} = \frac{3}{5}$$

$$b = y_0 - m \times x_0 \\ = 2 - \frac{3}{5} \times 2$$

$$= \frac{4}{5}$$

$$y = mx + b$$

$$y(3) = \frac{3}{5} \times 3 + \frac{4}{5} = 2 \frac{3}{5} = 2.6 \approx 3$$

$$y(4) = \frac{3}{5} \times 4 + \frac{4}{5} = 3 \frac{1}{5} = 3.2 \approx 3$$

$$y(5) = \frac{3}{5} \times 5 + \frac{4}{5} = 3 \frac{4}{5} = 3.8 \approx 4$$

$$y(6) = \frac{3}{5} \times 6 + \frac{4}{5} = 4 \frac{2}{5} = 4.4 \approx 4$$

mishandle,

$$(2, 2) \rightarrow (5, 9)$$

$$y(3) = 4.33 \approx 4$$

$$y(4) = 6.67 \approx 7$$

$$m = \frac{7}{3}$$

$$b = -\frac{8}{3}$$

looks
(DISCONTINUOUS)

(Line drawing in the pixels is not exactly accurate)

A simple solⁿ (cont...)

* As m not in the range $-1 < m < 1$

so the problem arise

So, instead of working out y values,

we will calculate x values for each y values

$$y = mx + b \rightarrow x = \frac{y - b}{m} \quad (\text{If } m \text{ value} > 1)$$

$$x(3) = \frac{3 + \frac{8}{3}}{\frac{7}{3}} = 2 \frac{3}{7} \approx 2$$

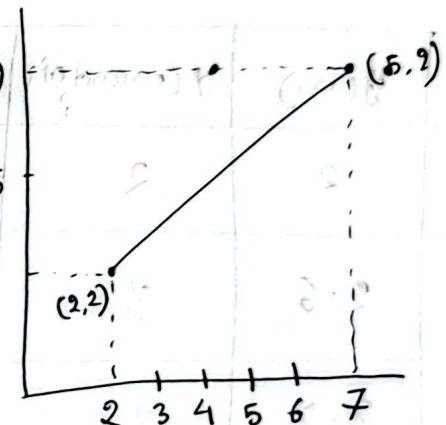
$$x(4) = \frac{4 + \frac{8}{3}}{\frac{7}{3}} = 2 \frac{6}{7} \approx 3$$

$$x(5) = \frac{5 + \frac{8}{3}}{\frac{7}{3}} = 3 \frac{2}{7} \approx 3$$

$$x(6) = \frac{6 + \frac{8}{3}}{\frac{7}{3}} = 3 \frac{5}{7} \approx 4$$

$$x(7) = \frac{7 + \frac{8}{3}}{\frac{7}{3}} = 4 \frac{1}{7} \approx 4$$

$$x(8) = \frac{8 + \frac{8}{3}}{\frac{7}{3}} = 4 \frac{4}{7} \approx 5$$



Pixel

(2, 3)

(3, 4)

(3, 5)

(4, 6)

(4, 7)

(5, 8)

{ NOT DISCONTINUOUS }

(x, y)
 $\swarrow$
 absissa $\rightarrow$ ordinate

The DDA Algorithm

(Used for removing multiplication term)

$$y_{k+1} = y_k + m$$

$$x_{k+1} = x_k + 1$$

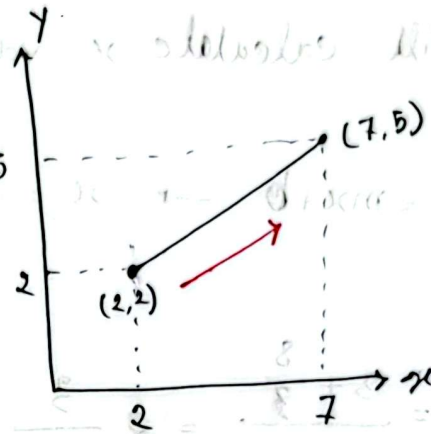
From prev $\rightarrow$

$$y(3) = 2.6$$

$$+1 \rightarrow y(4) = 3.2$$

$$+1 \rightarrow y(5) = 3.8$$

$$+1 \rightarrow y(6) = 4.4$$



x_{k+1}	y_{k+m}	$y(\text{round off})$	Pixel
2	2	2	(2, 2)
3	2.6	3	(3, 3)
4	3.2	3	(4, 3)
5	3.8	4	(5, 4)
6	4.4	4	(6, 4)

$$m = \frac{5 - 2}{7 - 2} = \frac{3}{5}$$

$$\therefore m = 0.6$$

(1 Multiply only)
 { SO NOT COSTLY }

Step - 1: Calculate m

$$m = \frac{y_1 - y_2}{x_1 - x_2} = \frac{dy}{dx}$$

Step-2: (Formula Define)

If: $-1 < m < 1$,

$$x_{k+1} = x_k \pm 1$$

$$y_{k+1} = y_k \pm |m|$$

Else:

$$y_{k+1} = y_k \pm 1$$

$$x_{k+1} = x_k \pm \frac{1}{|m|}$$

Step-3: (Sign determine)

$$dx > 0 \rightarrow x+$$

$$dx \leq 0 \rightarrow x-$$

$$dy > 0 \rightarrow y+$$

$$dy \leq 0 \rightarrow y-$$

$$\square (-1011, -2022), (-2022, -1011)$$

$$\text{sl: } m = \frac{-1011 + 2022}{-2022 + 1011} = \frac{1011}{-1011} = -1 \quad (\text{Close})$$

$$\text{So, } x_{k+1} = x_k \pm 1$$

$$y_{k+1} = y_k \pm m = y_k \pm 1$$

$$\text{then, } x_{k+1} = x_k - 1 \quad (dx \text{ -ve so -1})$$

$$y_{k+1} = y_k + 1 \quad (dy \text{ +ve so +1})$$

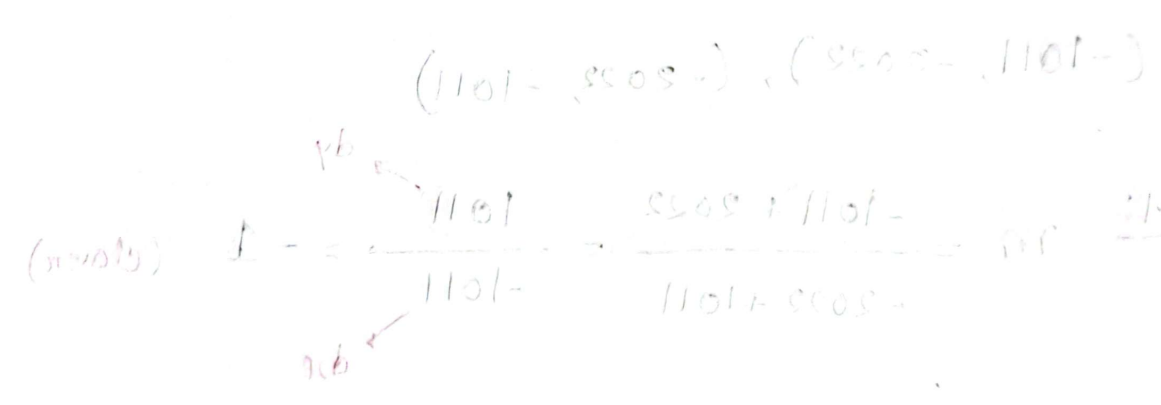
$x(-1)$	$y(+1)$	$y(\text{round off})$	Pixel
-1011	-2022	-2022	(-1011, -2022)
-1012	-2021	-2021	(-1012, -2021)
-1013	-2020	-2020	(-1013, -2020)
-1014	-2019	-2019	(-1014, -2019)
-1015	-2018	-2018	(-1015, -2018)

↓ ↓ ↓ ↓
↓ ↓ ↓ ↓

→ DDA Algorithm solves **Multiplication problem** but it doesn't solve **floating point** problem

(So Not a Perfect Algorithm)

$$\begin{array}{l|l}
 x_b \leftarrow 0 < x_b & x_b \leftarrow 0 < x_b \\
 -x_b \leftarrow 0 > x_b & -x_b \leftarrow 0 > x_b
 \end{array}$$



x_b	y_b	x_b	y_b
101	505	505	101

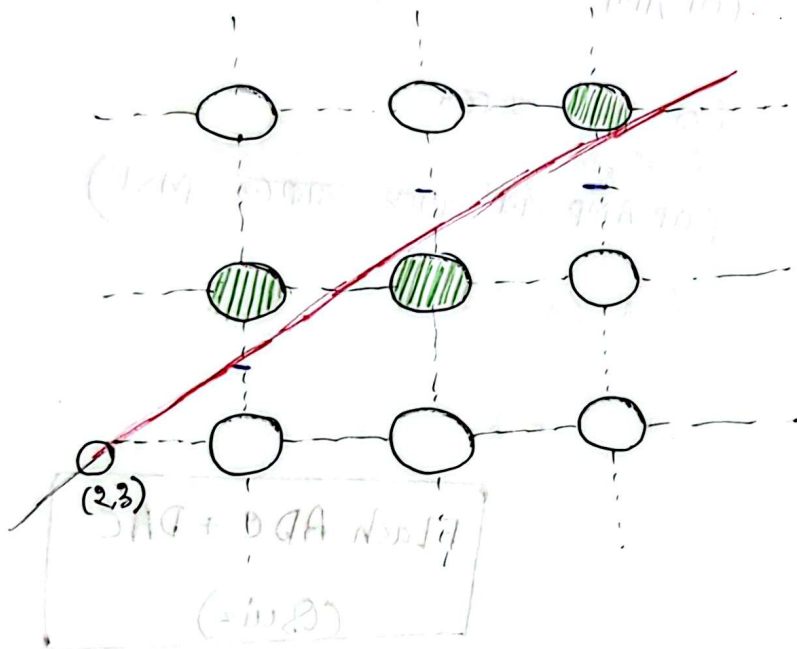
Midpoint Line Algorithm (MPL)

In DDA m is float num,

So problem occurs

(i) Add vs Mul

(ii) Int vs Float



* $5x - 3y - 1 = 0$ (straight line equation)

$$f(2,3) = 5 \times 2 + 3 \times 3 - 1 = 0$$

$$f(2,4) = 5 \times 2 - 3 \times 4 - 1 < 0$$

(lower pixel choose)

$$f(2,5) = 5 \times 2 - 3 \times 5 - 1 = -9 > 0$$

(upper pixel choose)

$f(M)$	Pixel chosen
$f(M) > 0$	upper (NE)
$f(M) \leq 0$	lower (E)

for one scenario

if $f(x,y) = 0$ choose any pixel but be consistent in choosing one. Hence we chose lower pixel

$$(1) 0 \leq m \leq 1$$

$$(2) \Delta x, \Delta y > 0$$

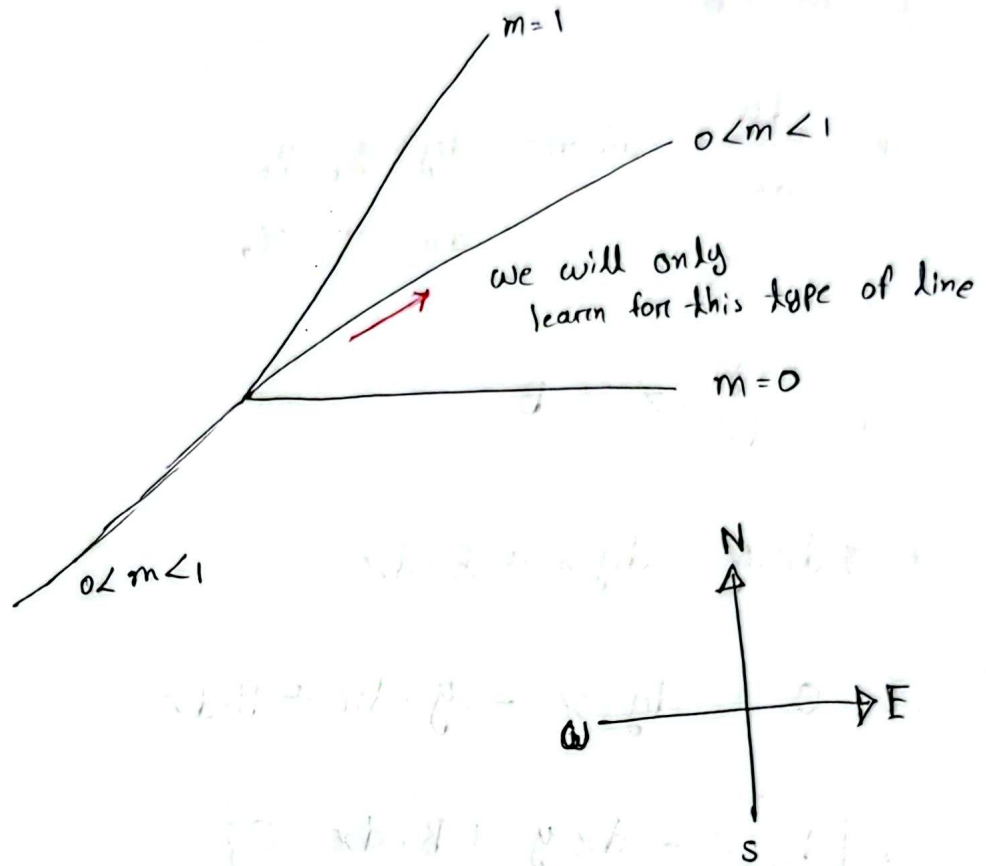
$$m = \frac{\Delta y}{\Delta x}$$

$$\Delta x = x_2 - x_1$$

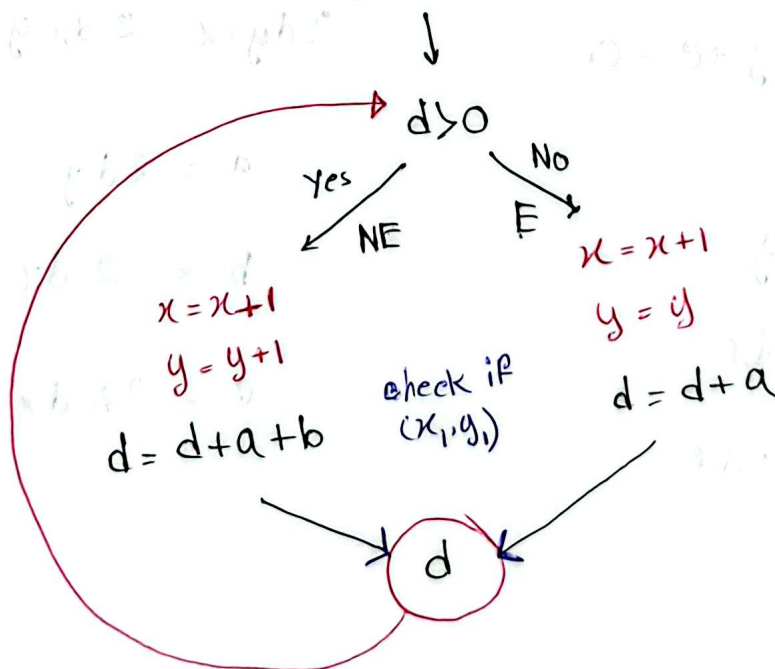
$$\Delta y = y_2 - y_1$$

$$\Delta x > 0$$

$$\Delta y > 0$$



initially $(x, y) \quad d = a + 0.5b$
 $d = f(M)$ Midpoint



So, $y = mx + B$

$m = \frac{dy}{dx}$ where, $dy = y_1 - y_0$
 $dx = x_1 - x_0$

$y = \frac{dy}{dx} x + B$

$\Rightarrow dx \cdot y = dy \cdot x + B \cdot dx$

$\Rightarrow 0 = dy \cdot x - y \cdot dx + B \cdot dx$

$\Rightarrow \boxed{dy \cdot x - dx \cdot y + B \cdot dx = 0}$

Comparing $\rightarrow$

$ax + by + c = 0$

we get,

$a = dy$

$b = -dx$

$c = B \cdot dx$

$\hookrightarrow \times 2$

$2dy \cdot x - 2dx \cdot y + 2B \cdot dx = 0$

$a = 2dy$

$b = -2dx$

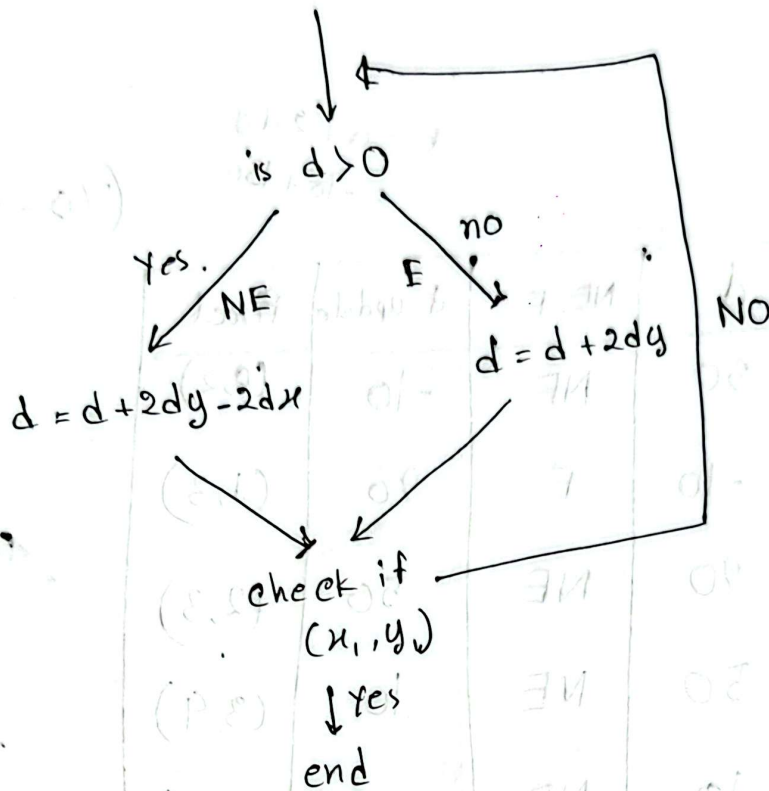
$c = 2B \cdot dx$

$$d = d_{\text{init}}$$

$$= 2dy - dx$$

$$d = a + 0.5b$$

$dx, dy \rightarrow \text{int}$



putting
 $a = 2dy$
 $b = -2dx$

$$2dy = dy + dy$$

$$2dx = dx + dx$$

Q. Find out the first 7 pixels of the line segment starting from (0,2) to (70,52) using midpoint line algorithm.

Ans:

$$E = d + 2dy$$

$$= -10 + 100$$

$$(70 - 0 - 1)$$

$$= 69 \text{ pixels}$$

x	y	d	NE, E	d update	Pixel
0	2	30	NE	-10	(0,2)
1	3	-10	E	90	(1,3)
2	3	90	NE	50	(2,3)
3	4	50	NE	10	(3,4)
4	5	10	NE	-30	(4,5)
5	6	-30	E	70	(5,6)
6	6	70	NE	30	(6,6)
7	7	30	NE	-10	(7,7)
8	8				

$$(0,2) \quad (70,52)$$

$$m = \frac{dy}{dx} = \frac{50}{70} < 1$$

$$\Delta x, dx > 0$$

$$\Delta y, dy > 0$$

$$d = 2dy - dx$$

$$= 100 - 70$$

$$= 30$$

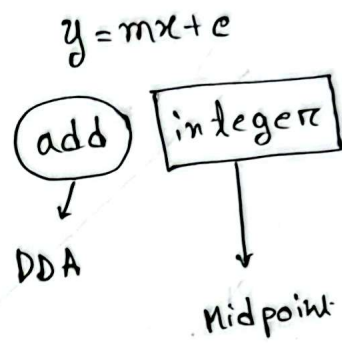
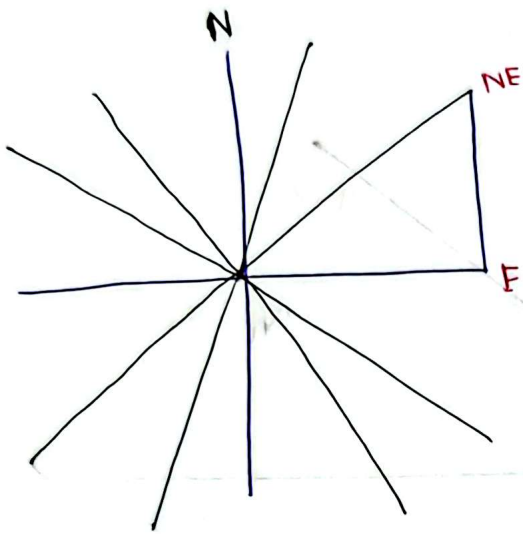
$$NE = 2(dy - dx)$$

$$= 40$$

$$d = d + 2dy - 2dx$$

$$= 30 + 2 \times 50 - 2 \times 70$$

Eight Way symmetry



for $0 < m < 1$

(iii) $\Delta y = +ve$
 $\Delta x = +ve$

{TOTAL LINE 8 TYPES}

condition:

(i) $0 < m < 1$

(ii) $\Delta x ++, \Delta y ++$

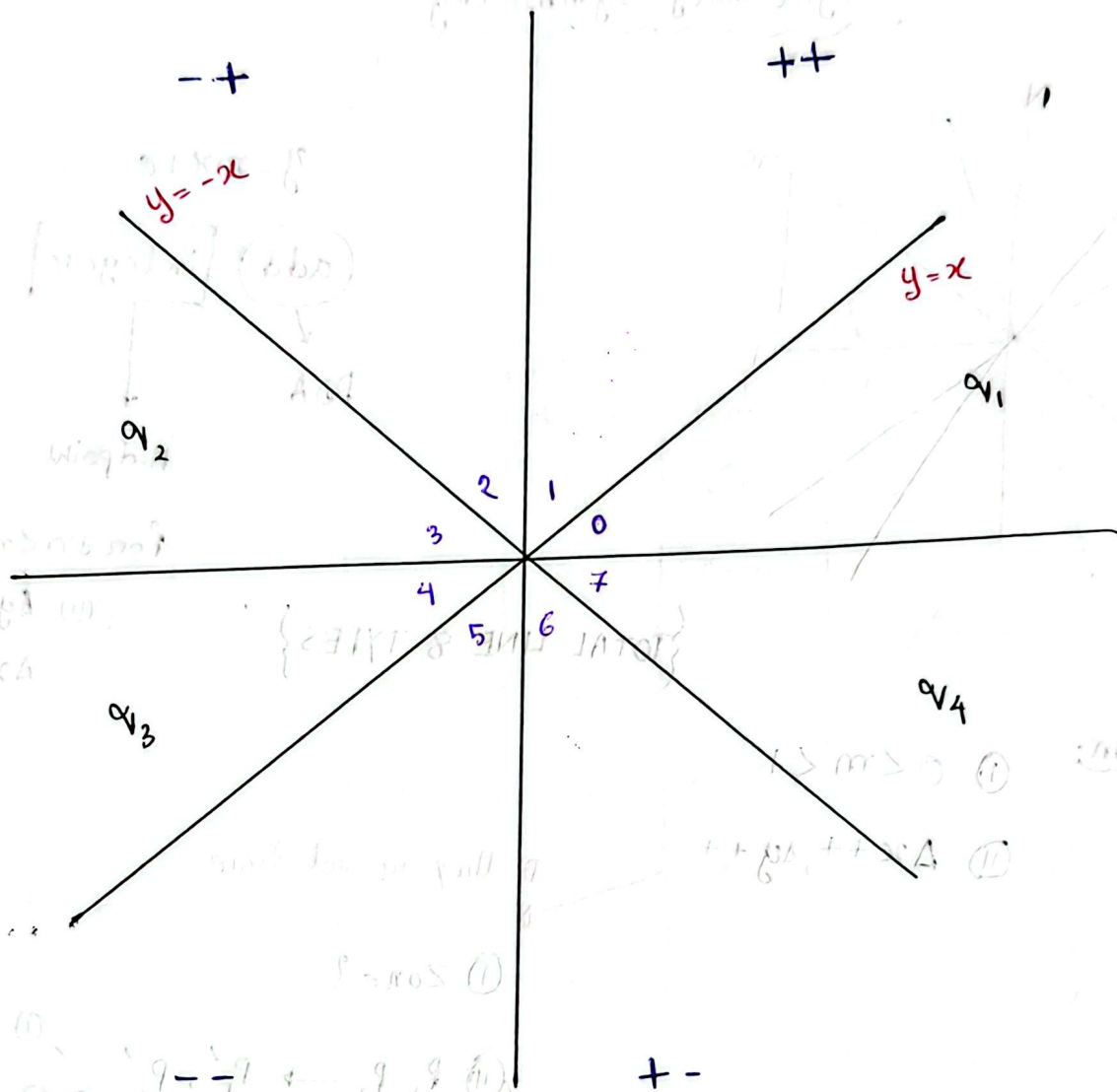
If they are not true

(i) Zone?

(ii) $P_1, P_2 \rightarrow P_1', P_2'$

(iii) Mid Point LDA $\rightarrow O_1, O_2, O_3$

(iv) $O_1, O_2 \rightarrow O_1', O_2'$



For Finding Zones:

$|dy| > |dx| \rightarrow$ closer to y axis

$|dx| > |dy| \rightarrow$ " " x "

$dy > 0, dx > 0 \rightarrow q_1$
 $dy > 0, dx < 0 \rightarrow q_2$
 $dy < 0, dx < 0 \rightarrow q_3$
 $dy < 0, dx > 0 \rightarrow q_4$

$P \rightarrow (x, y)$

$x > 0, y > 0 - q_1$

$x < 0, y > 0 - q_2$

(fix) P_1
 (infinity) P_2
 (x_1, y_1) (x_2, y_2)

$$dy = y_2 - y_1$$

$$dx = x_2 - x_1$$

$$\square (-50, -15), (100, 35)$$

$$dx = 100 + 50 = 150 > 0$$

$$dy = 35 + 15 = 50 > 0$$

$$|dx| > |dy|, \text{ Zone} = 0$$

Direct use Midpoint
line drawing.

Zone Converting:

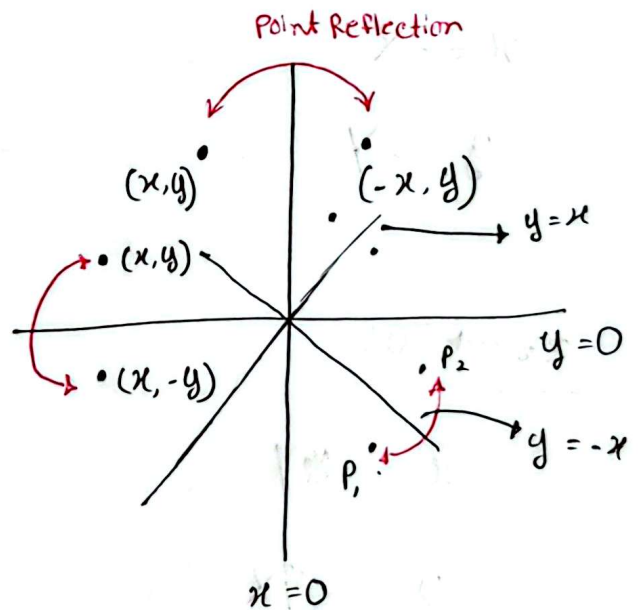
$$\textcircled{i} (x, y) \xrightarrow{x=0} (-x, y)$$

$$\textcircled{ii} (x, y) \xrightarrow{y=0} (x, -y)$$

$$\textcircled{iii} (x, y) \xrightarrow{y=x} (y, x)$$

$$\textcircled{iv} (x, y) \xrightarrow{y=-x} (-y, -x)$$

Step-1: Convert Every zone
to zone 0



Input: x, y Output: x', y'

$z1 \rightarrow z0:$ $x = y, y = x$

$z2 \rightarrow z0:$

$x = y$

$y = -x$

$z3 \rightarrow z0:$

$x = -x$

$y = y$

$z4 \rightarrow z0:$

$x = -x$

$y = -y$

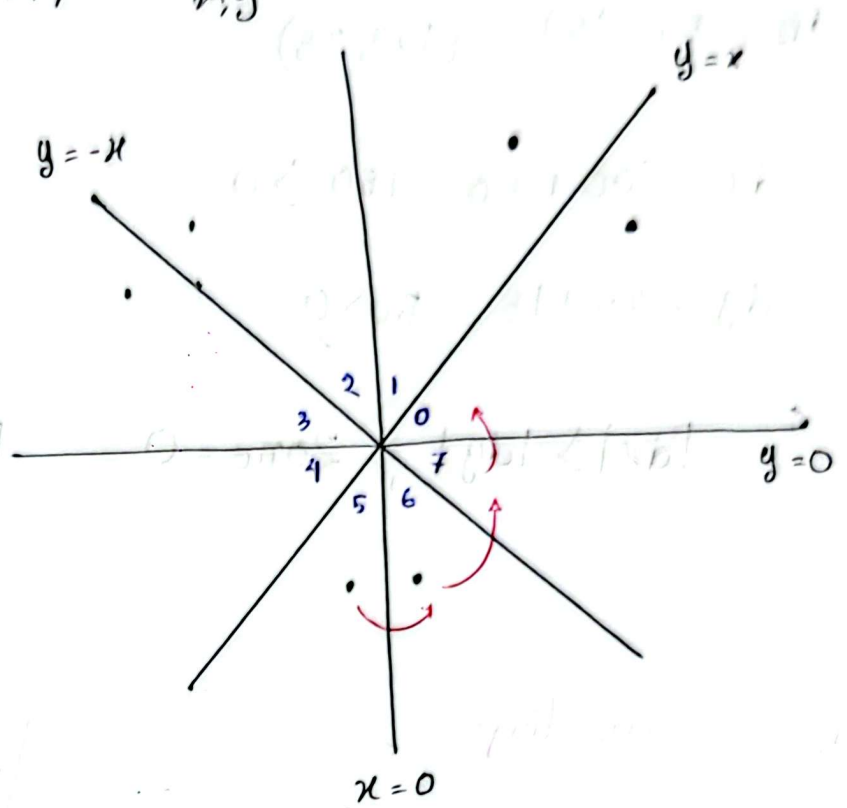
$z5 \rightarrow z0:$

$x = -y,$

$y = -x$

$z6 \rightarrow z0:$

$x = -y, y = x$



step-2

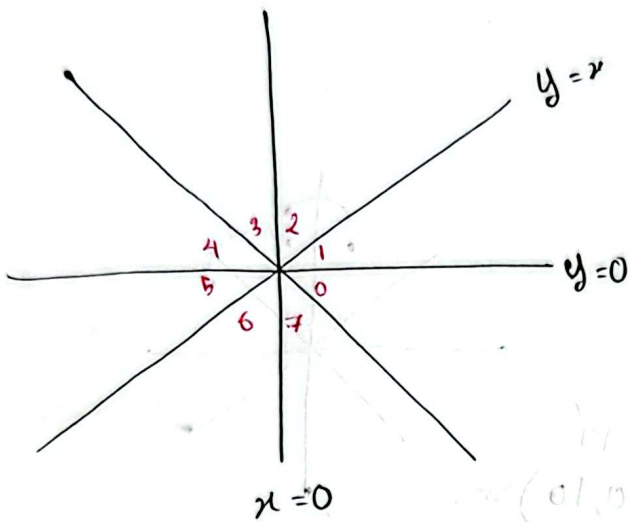
(Here, we are

converting to
zone 0)

$z7 \rightarrow z0:$

$x = x,$

$y = -y$



step 4

Back to the original
zone

$$\underline{z0 - z1:}$$

$$x = y$$

$$y = x$$

$$\underline{z0 - z2:}$$

$$x = -y$$

$$y = -x$$

$$\underline{z0 - z3:}$$

$$x = -x$$

$$y = y$$

$$\underline{z0 - z4:}$$

$$x = -x,$$

$$y = -y$$

$$\underline{z0 - z5:}$$

$$x = -y$$

$$y = -x$$

$$\underline{z0 - z7:}$$

$$x = x$$

$$y = -y$$

$$\underline{z0 - z6:}$$

$$x = y$$

$$y = -x$$

□ $(-10, -20)$ to $(-20, 70)$

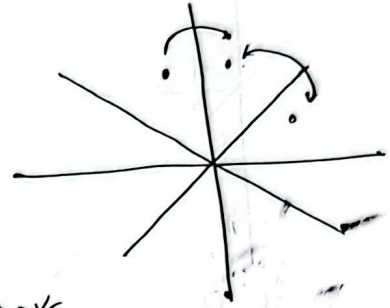
$$dx = -20 + 10 = -10 < 0$$

$$dy = 70 + 20 = 90 > 0$$

$$|dy| > |dx|, \text{ zone} = 2$$

$$\overset{P_1}{(-10, -20)} \rightarrow \overset{P_1'}{(10, -20)} \rightarrow \overset{P_1'}{(-20, 10)} \swarrow$$

$$\overset{P_2}{(-20, 70)} \rightarrow \overset{P_2'}{(20, 70)} \rightarrow \overset{P_2'}{(70, 20)} \swarrow$$



Back to the original

some